2201NSC Linear Algebra and Applications

Week 8

David Harman

Matrix Factorisations

PLU, QR, Cholesky and SVD

LU factorisation records Gaussian elimination

$$A = LU$$

- U is upper triangular (or upper trapezoidal) and can be obtained by applying row operations to A until it is in row-echelon form.
- L is lower triangular (or lower trapezoidal) and records the reverse of the row operations needed to recover A from U.

The limitation

An LU factorisation is not always possible. If elimination requires a row swap to obtain a nonzero pivot, that swap changes the order of the rows and cannot be recorded by a lower-triangular matrix L. A permutation matrix is then required, leading to a PLU factorisation.

PLU factorisation includes the required row swaps

When row swaps are required, introduce a permutation matrix P:

$$A = PLU.$$

- P records the row permutations;
- L records the row replacements;
- U is the resulting upper-triangular or upper-trapezoidal matrix.

PLU can be applied to rectangular as well as square matrices. The dimensions of the factors depend on the chosen full or reduced form.

MATLAB convention

MATLAB's `[L,U,P]=lu(A)` returns $PA = LU$. Therefore $A = P^{\mathsf{T}}LU$ in the convention used here.

QR factorisation gives orthonormal column directions

$$A = QR$$

The columns of Q are orthonormal, so

$$Q^{\mathsf{T}} Q = I.$$

The upper-triangular or upper-trapezoidal matrix R contains the coordinates needed to reconstruct the columns of A from those directions.

Construction

Gram–Schmidt orthogonalisation can construct Q from the columns of A. Numerical software usually uses more stable methods.

Full and thin QR contain different numbers of columns

Let $A \in \mathbb{R}^{m \times n}$ have rank r.

Full QR

$$A = QR, \quad Q \in \mathbb{R}^{m \times m}.$$

Q is square and orthogonal. Extra columns complete an orthonormal basis for $\mathbb{R}^m$.

Thin QR

$$A = QR, \quad Q \in \mathbb{R}^{m \times r}, \quad R \in \mathbb{R}^{r \times n}.$$

Also called the **compact** or **rank-reduced** QR factorisation.

In both forms, the columns of Q are orthonormal and $Q^\mathsf{T} Q = I$.

Cholesky uses one triangular factor

If A is real, symmetric and positive definite, then

$$A = U^{\mathsf{T}} U \qquad \text{or equivalently} \qquad A = LL^{\mathsf{T}},$$

where U is upper triangular and $L = U^{\mathsf{T}}$ is lower triangular.

Requirements

- square;
- symmetric;
- positive definite.

Benefits

Uses less storage and work than a general LU factorisation, while preserving symmetry.

SVD applies to every real matrix

For any $A \in \mathbb{R}^{m \times n}$,

$$A = U \Sigma V^{\mathsf{T}},$$

where

- $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal;
- $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal;
- $\sigma_1, \sigma_2, \ldots$ are the diagonal entries of Σ, called the **singular values**, and are ordered so that $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

The singular values reveal the strength of independent directions in the transformation. The number of nonzero singular values is $\operatorname{rank}(A)$.

The thin SVD retains exactly the rank information

If $A \in \mathbb{R}^{m \times n}$ has rank r, its thin SVD is

$$A = U \Sigma V^{\mathsf{T}},$$

with

$$U \in \mathbb{R}^{m \times r}, \qquad \Sigma \in \mathbb{R}^{r \times r}, \qquad V \in \mathbb{R}^{n \times r}.$$

It is also called the **compact** or **rank-reduced** SVD.

The columns of U and V are orthonormal, although the matrices are usually not square.

SVD connects a matrix to four fundamental subspaces

For $A \in \mathbb{R}^{m \times n}$ with rank r, using the *full* SVD:

Columns of U

- First r: basis for $\mathrm{Col}(A)$.
- Final $m - r$: basis for $\mathcal{N}(A^\mathsf{T})$.

Columns of V

- First r: basis for $\mathrm{Col}(A^\mathsf{T})$.
- Final $n - r$: basis for $\mathcal{N}(A)$.

Also,

$$A^\mathsf{T} A = V \Sigma^\mathsf{T} \Sigma V^\mathsf{T}, \qquad A A^\mathsf{T} = U \Sigma \Sigma^\mathsf{T} U^\mathsf{T}.$$

Thus the squared singular values are the nonzero eigenvalues of both matrices.

Summary of the four factorisations

Method	Form	When and why	MATLAB
PLU	$A = PLU$	General square or rectangular matrices; records Gaussian elimination and row swaps.	`[L,U,P]=lu(A)`
QR	$A = QR$	General square or rectangular matrices; gives orthonormal column directions and supports least squares.	`[Q,R]=qr(A)`
Cholesky	$A = U^{\top} U$	Symmetric positive-definite square matrices; uses one triangular factor.	`U=chol(A)`
SVD	$A = U \Sigma V^{\top}$	Any square or rectangular matrix; reveals rank and fundamental subspaces.	`[U,S,V]=svd(A)`

Important conventions

MATLAB's LU output satisfies $PA = LU$. In SVD, U and V are orthogonal, not triangular.

Short Revision Quiz

PLU, QR, Cholesky and SVD

Question 1 — LU notation

Which expression represents an LU factorisation?

A. $A = LU$

B. $A = L + U$

C. $A = UL^{\mathsf{T}}$

Question 2 — QR notation

In a QR factorisation, what kind of matrix is R?

A. Upper triangular or upper trapezoidal

B. Orthogonal

C. A permutation matrix

Question 3 — PLU notation

In a PLU factorisation, what is the role of P?

A. It records row permutations

B. It contains the eigenvalues

C. It orthogonalises the columns

Question 4 — LU structure

How is U obtained in an LU factorisation?

A. By row reducing A to row-echelon form

B. By finding the eigenvectors of A

C. By transposing A

Question 5 — Limitation of LU

Why might a matrix fail to have an LU factorisation?

A. Elimination requires a row swap

B. The matrix has positive entries

C. The matrix is rectangular

Question 6 — General applicability

Which factorisation is guaranteed to exist for every real matrix?

A. LU

B. Cholesky

C. SVD

Question 7 — Rectangular matrices

Which factorisation requires A to be square?

A. PLU

B. QR

C. Cholesky

Question 8 — Cholesky conditions

Which matrix is guaranteed to have a Cholesky factorisation?

A. Every square matrix

B. Every symmetric matrix

C. Every real symmetric positive-definite matrix

Question 9 — Cholesky forms

If $A = U^\top U$ is a Cholesky factorisation and $L = U^\top$, which equivalent form is correct?

A. $A = L^\top L$

B. $A = LL^\top$

C. $A = L + L^\top$

Question 10 — Singular values

Where do the singular values $\sigma_1, \sigma_2, \ldots$ appear in the SVD?

A. On the diagonal of Σ

B. On the diagonal of U

C. In the final row of $V^\top$

Question 11 — Signs of singular values

Which statement about singular values is correct?

A. They must all be negative

B. They are nonnegative and conventionally listed in descending order

C. They must all equal 1

Question 12 — Rank from the SVD

How is the rank of A identified from its singular values?

A. Count the negative singular values

B. Count the nonzero singular values

C. Add all the singular values

Question 13 — Full SVD dimensions

If $A \in \mathbb{R}^{5\times 3}$, what are the dimensions of U and V in the full SVD?

A. U is 5×5 and V is 3×3

B. U is 5×3 and V is 3×3

C. U is 3×3 and V is 5×5

Question 14 — Thin QR dimensions

Let $A \in \mathbb{R}^{5\times 3}$ have rank 3. In its thin QR factorisation, what are the dimensions of Q and R?

A. Q is 5×5 and R is 5×3

B. Q is 5×3 and R is 3×3

C. Q is 3×3 and R is 3×5

Question 15 — SVD notation

In the SVD
$$A = U\Sigma V^{\top},$$
what kind of matrix is U?

A. Upper triangular or upper trapezoidal

B. Orthogonal

C. Diagonal

Quiz answers 1–5

1 — A

An LU factorisation has the form $A = LU$.

2 — A

R is simply another conventional symbol for an upper-triangular factor.

3 — A

P records the row swaps required during elimination.

4 — A

Row operations reduce A to the row-echelon matrix U.

5 — A

A required row swap cannot be recorded by a lower-triangular L alone.

Quiz answers 6–10

6 — C

Every real matrix has an SVD.

7 — C

Cholesky requires a square matrix; PLU and QR also apply to rectangular matrices.

8 — C

Real, symmetric and positive definite are the Cholesky conditions.

9 — B

If $L = U^\top$, then $U^\top U = LL^\top$.

10 — A

The singular values are the diagonal entries of Σ.

Quiz answers 11–15

11 — B

Singular values are nonnegative and are ordered from largest to smallest.

12 — B

The number of nonzero singular values equals the rank.

13 — A

In the full SVD, $U \in \mathbb{R}^{5\times 5}$ and $V \in \mathbb{R}^{3\times 3}$.

14 — B

For rank 3, thin QR has $Q \in \mathbb{R}^{5\times 3}$ and $R \in \mathbb{R}^{3\times 3}$.

15 — B

Despite the letter used, U in the SVD is orthogonal, not upper triangular or upper trapezoidal.